

# Deep Reinforcement Learning

## *Value function approximation*

Prof. Dr. Sebastian Peitz

Chair of Safe Autonomous Systems, TU Dortmund

Summer term 2026

# Content

# Content

- Limitations of tabular methods
- Value estimation / prediction with function approximation
- Gradient-based prediction
- Batch learning

# Where are we?

| Chapter | Topic                                                                  | Content                                      |
|---------|------------------------------------------------------------------------|----------------------------------------------|
|         | <b>Basics &amp; tabular methods</b>                                    |                                              |
| 1-5     | Bandits, MDPs, Dynamic Programming, Monte Carlo, TD Learning           | RL basics in finite dimensions               |
|         | <b>Deep-learning-based methods</b>                                     |                                              |
| 6       | Brief introduction to deep learning                                    | The basics for what comes next               |
| 7       | Value function approximation                                           | Value estimation with function approximation |
| 8       | Deep $Q$ -learning                                                     |                                              |
| 9       | Policy gradients                                                       |                                              |
| 10      | Actor-critic algorithms                                                |                                              |
| 11      | Advanced algorithms (Part I): From policy gradient to PPO              |                                              |
| 12      | Advanced algorithms (Part II): From $Q$ -learning to Soft Actor-Critic |                                              |
|         | <b>Model-Based Control</b>                                             |                                              |
|         | <b>Advanced Topics</b>                                                 |                                              |

# Limitations of tabular methods

# Limitations of tabular methods

We have essentially “solved” the RL problem in part one of our lecture! But there are quite a few limitations when it comes to real-world systems:

- The **curse of dimensionality**: For large dimensions  $|\mathcal{S}|$  or  $|\mathcal{A}|$  (even moderate ones), the calculations become prohibitively expensive and convergence is very slow.
- **Continuous state spaces**, i.e., infinitely many possible states

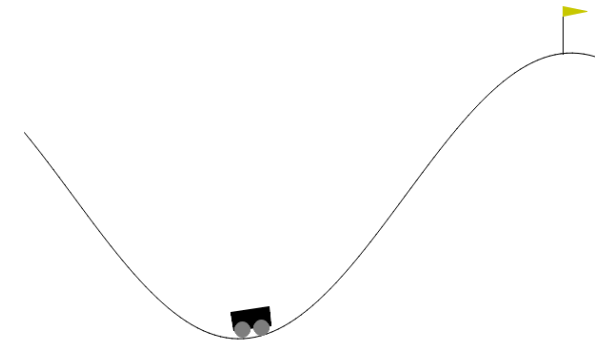
$$s \in \mathcal{S} \subseteq \mathbb{R}^n \quad / \quad |\mathcal{S}| = \infty. \quad (\text{instead of } \mathcal{S} = \{s_1, \dots, s_{|\mathcal{S}|}\})$$

- **Example**: The *mountain car*, with continuous  $\mathcal{S} = [x_{\min}, x_{\max}] \times [v_{\min}, v_{\max}]$  (minimal and maximal position  $x$  and velocity  $v$ ), but finite  $\mathcal{A} = \{-1, 0, 1\}$  (accelerate left, do nothing, accelerate right).

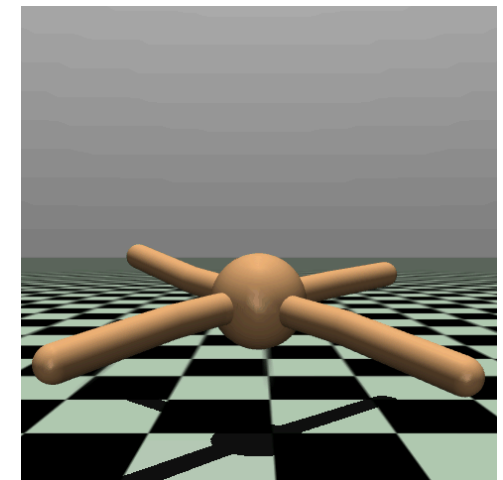
- **Continuous action spaces**, i.e., infinitely many possible actions\*

$$a \in \mathcal{A} \subseteq \mathbb{R}^m \quad / \quad |\mathcal{A}| = \infty. \quad (\text{instead of } \mathcal{A} = \{a_1, \dots, a_{|\mathcal{A}|}\})$$

- **Example**: The *ant*, with 27 continuous state variables and 8 continuous action variables.



Mountain car [Source].



\* For now, however, we will stick to finite action spaces.



# Value estimation / prediction with function approximation



# Value estimation / prediction with function approximation

- From now on, we will introduce function approximators for the quantities of interest.
- For the value function, this means that we're looking for

$$V_{\theta}(s) \approx V^{\pi}(s).$$

- $V_{\theta}(s)$  may be a *linear model*, i.e.,

$$V_{\theta}(s) = \sum_{k=1}^d \theta_k \psi_k(s) = \theta_1 \psi_1(s) + \dots + \theta_d \psi_d(s),$$

- but also a *nonlinear model* such as a deep neural network.
- The weight vector  $\theta \in \mathbb{R}^d$  typically is of lower dimension  $d$  than the state space  $\mathcal{S}$ :  $d \ll |\mathcal{S}|$ .
  - Otherwise this exercise would be pointless!
  - This obviously holds for continuous state spaces, but also for very large yet finite-dimensional state spaces.

## A note on the state space dimension

We use  $n$  to denote the number of state variables, but this does not mean that we need to have  $n = \infty$  for continuous state spaces. A state space is finite if each state variable  $s_k$  is an element of a finite set, whereas it is continuous, if at least one  $s_1, \dots, s_n$  is a continuous variable.

## Global impact of experience

Updates due to new experience lead to updates in our model parameter  $\theta$ . This means that *we modify our value estimate for many states*, in contrast to tabular methods where we update individual states only.

# Prediction objective

- In the tabular case a specific prediction objective was not needed:
  - The learned value function could exactly match the true value.
  - The value estimate at each state was decoupled from other states.
- Due to global impact of experience, we need to define an accuracy metric on the entire state space: the **RL prediction goal**.

## Definition: Value error (VE)

The RL prediction objective is defined as the mean squared value error

$$\overline{VE}(\theta) = \int_{\mathcal{S}} \mu(s) \left( V^{\pi}(s) - V_{\theta}(s) \right)^2 ds.$$

Here,  $\mu(s) \geq 0$  is the *state distribution*, with  $\int_{\mathcal{S}} \mu(s) ds = 1$ .

## Challenge:

- The true value  $V^{\pi}(s)$  is unknown in most tasks!
- We need to find ways to estimate  $V^{\pi}(s)$  from experience.

# The on-policy prediction objective

- For prediction we will **focus entirely on the on-policy case**.
- Hence,  $\mu(s)$  is the on-policy distribution under the current policy  $\pi$ .

## On policy prediction objective

As  $\overline{VE}(\theta)$  is the expected error, we can use *Monte Carlo sampling* to estimate it:

$$L(\theta) = \sum_{k=1}^N \left( V^{\pi}(s_k) - V_{\theta}(s_k) \right)^2 \approx \overline{VE}(\theta). \quad (1)$$

**Goal:** find the optimal value estimate  $V_{\theta^*}$  by optimization:

$$\theta^* = \arg \min_{\theta} L(\theta).$$

## Challenges:

1. The true value  $V^{\pi}(s)$  is unknown in most tasks!
2. The function approximator  $V_{\theta}(s)$  needs to be able to approximate  $V^{\pi}(s)$ .
3. Training:
  - If  $V_{\theta}(s)$  is a linear model: convex optimization problem.
    - The “easy” case: local optimum = global optimum, uniquely defined.
  - If  $V_{\theta}(s)$  is nonlinear model: nonlinear optimization problem.
    - The “hard” case: many local optima with no guarantee to locate the global one.
    - Training may diverge, advanced optimization techniques are very important!

💡 **Off-policy prediction:** *importance sampling* to transform the sampled value estimates from the behavior to the target policy.  
⇒ Increases the prediction variance; in combination with generalization errors due to function approximation: large risk of diverging.

# Gradient-based prediction

# Incremental weight updates

- Now that we have an objective  $L(\theta)$  we would like to optimize for (Eq. (1)), let's use **gradient descent** with learning rate  $\alpha$ :

$$\theta^{(t+1)} = \theta^{(t)} - \frac{1}{2}\alpha \nabla L(\theta^{(t)}) = \theta^{(t)} - \frac{1}{2}\alpha \nabla_{\theta} \left( \sum_{k=1}^N (V^{\pi}(s_k) - V_{\theta}(s_k))^2 \right).$$

- What's the problem here?  
 $\Rightarrow$  We often only have a *single sample*. (Or, instead, we would like to perform an update after a single sample.)

## SGD-based parameter update

Given a new sample  $s_t$  and its (for now exact) value  $V^{\pi}(s_t)$ , we can update our current weight vector  $\theta^{(t)}$  by using **stochastic gradient descent (SGD)** with:

$$\theta^{(t+1)} = \theta^{(t)} - \frac{1}{2}\alpha \nabla_{\theta} [(V^{\pi}(s_t) - V_{\theta}(s_t))^2] = \theta^{(t)} + \alpha [V^{\pi}(s_t) - V_{\theta}(s_t)] \nabla_{\theta} V_{\theta}(s_t). \quad (2)$$

**Question:** Should we follow (2) to optimality?

$\Rightarrow$  No! We are optimizing for a single sample  $\rightarrow$  **overfitting**.

$\Rightarrow$  Take a single, small step only, then collect new experience.

# Bootstrapping the true value

Challenge 1. remains: We do not know  $V^\pi(s_t)$  which we need for our prediction objective (1).

What can we do?  $\Rightarrow$  Use an estimate for  $V^\pi(s_t)$  that we can compute from experience!

**Possible options for estimates:**

## Unbiased estimates of $V^\pi(s_t)$

Monte Carlo estimates for  $g_t$ , sample from entire trajectories.

## Bootstrapped estimates of $V^\pi(s_t)$

- The Dynamic Programming target

$$V_\theta(s_t) \leftarrow \sum_{a \in \mathcal{A}} \pi(a|s_t) \sum_{s' \in \mathcal{S}} p(s'|s_t, a) [r + \gamma V_\theta(s')].$$

- TD targets (e.g., TD(0))

$$V_\theta(s_t) \leftarrow V_\theta(s_t) + \alpha [r_t + \gamma V_\theta(s_{t+1}) - V_\theta(s_t)].$$

💡 Inserting the bootstrapped versions into (2) does not yield a true gradient descent method, as we are ignoring the parametric dependency of the target on  $\theta$  (Sutton and Barto 1998). More on this on the next slides.

# Algorithm: Gradient Monte Carlo value estimation

Algorithm: Gradient Monte Carlo algorithm for estimating  $V_\theta \approx V^\pi$

Input:

- The current policy  $\pi$
- a differentiable, parameter-dependent function  $V_\theta : \mathcal{S} \rightarrow \mathbb{R}$
- learning rate  $\alpha$

Initialize: Value function weights  $\theta \in \mathbb{R}^d$  arbitrarily

for  $k = 1, 2, \dots, K$  episodes:

    Generate a sequence following  $\pi$ :

$$((s_0, a_0, r_0), (s_1, a_1, r_1), \dots, (s_{T_k-1}, a_{T_k-1}, r_{T_k-1}))$$

    Calculate the every-visit returns  $g_t$

    for  $t = 0, 1, \dots, T - 1$ :

$$\theta \leftarrow \theta + \alpha [g_t - V_\theta(s_t)] \nabla_\theta V_\theta(s_t)$$

- Direct transfer from the tabular case to function approximation.

# Semi-gradient methods

- If bootstrapping is applied, the true target  $V^\pi(s_t)$  is approximated by a target depending on the estimate  $V_\theta(s_t)$ .
- If  $V_\theta(s_t)$  does not fit  $V^\pi(s_t)$  (which it does not before convergence), then the update target becomes a *biased estimate*.
  - For example, in the TD(0) case, we get for (1):

$$L(\theta) = \sum_{t=1}^T (r_t + V_\theta(s_{t+1}) - V_\theta(s_t))^2.$$

- Taking the gradient of the single-sample version for  $L(\theta)$  yields:

$$\begin{aligned}\theta &\leftarrow \theta - \frac{1}{2} \alpha \nabla_\theta [(r_t + \gamma V_\theta(s_{t+1}) - V_\theta(s_t))^2] \\ &= \theta - \alpha [r_t + \gamma V_\theta(s_{t+1}) - V_\theta(s_t)] \nabla_\theta [\gamma V_\theta(s_{t+1}) - V_\theta(s_t)] \\ &\neq \theta + \alpha [r_t + \gamma V_\theta(s_{t+1}) - V_\theta(s_t)] \nabla_\theta V_\theta(s_t) \quad (\text{Eq. (2)}).\end{aligned}$$

- Application of Eq. (2) is still very common.
- These approaches are known as **semi-gradients** because they do not approximate the actual gradient.

## TD(0) semi-gradient update of $\theta$

$$\theta \leftarrow \theta + \alpha [r_t + \gamma V_\theta(s_{t+1}) - V_\theta(s_t)] \nabla_\theta V_\theta(s_t). \quad (3)$$



# Algorithm: Semi-gradient TD(0) value estimation

**Algorithm: Semi-gradient TD(0) for estimating  $V_\theta \approx V^\pi$**

*Input:*

- The current policy  $\pi$
- a differentiable, parameter-dependent function  $V_\theta : \mathcal{S} \rightarrow \mathbb{R}$
- learning rate  $\alpha$

*Initialize:* Value function weights  $\theta \in \mathbb{R}^d$  arbitrarily

**for**  $k = 1, 2, \dots, K$  episodes:

    Initialize  $s_0$

**for**  $t = 0, 1, \dots, T - 1$ :

        Apply action  $a_t \sim \pi(\cdot | s_t)$

        Observe  $r_t$  and  $s_{t+1}$

$\theta \leftarrow \theta + \alpha[r_t + \gamma V_\theta(s_{t+1}) - V_\theta(s_t)] \nabla_\theta V_\theta(s_t)$

**if**  $s_{t+1}$  is terminal **then STOP**

**Note:  $n$ -step version TD( $n$ )**

The above algorithm can be extended towards  $n$ -step bootstrapping by replacing the TD(0) target  $r_t + \gamma V_\theta(s_{t+1})$  with the  $n$ -step bootstrapped return

$$g_{t:t+n} = r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \dots + \gamma^{n-1} r_{t+n-1} + \gamma^n V_\theta(s_{t+n}).$$

The most challenging part in the implementation is the assessment of the remaining number of steps (i.e., when  $T - t < n$ ).

For details, see (Sutton and Barto 1998, Ch. 9.4).

# Batch learning

# Batch learning / experience replay

- As discussed during the TD learning lecture, training on individual samples can be slow.
- This also applies to SGD-based updates with bootstrapping.
- Alternative: Batch learning methods

## Mini-batch semi-gradient TD(0) with experience replay

Based on a dataset  $\mathcal{D} = \{(s_0, V^\pi(s_0)), \dots, (s_N, V^\pi(s_N))\}$ , repeat

- Sample uniformly  $b$  state-value pairs from experience (so-called mini batches), i.e.,  $\mathcal{B} = \{(s_i, V^\pi(s_i))\}_{i=1}^b \subset \mathcal{D}$ :

$$(s_i, V^\pi(s_i)) \sim \mathcal{D}.$$

- Apply (semi) SGD update step:

$$\theta \leftarrow \theta + \frac{\alpha}{b} \sum_{(s_i, V^\pi(s_i)) \in \mathcal{B}} [V^\pi(s_i) - V_\theta(s_i)] \nabla_\theta V_\theta(s_i).$$

## Remarks:

- Universally applicable:  $V_\theta$  can be any differentiable function.

- The true target  $V^\pi$  is usually approximated by MC or TD targets.

# Least squares policy evaluation

# Linear modeling of value functions

- Let's assume that we choose an approximation that is *linear* w.r.t. the weights  $\theta$ :

$$V_{\theta}(s) = \theta^{\top} \psi(s) = \sum_{k=1}^d \theta_k \psi_k(s).$$

- What's the gradient in this case?  $\Rightarrow$  it's simply the feature vector,

$$\nabla_{\theta} V_{\theta}(s) = \nabla_{\theta} \theta^{\top} \psi(s) = \psi(s).$$

- Our semi-gradient TD(0) update (3) thus simply becomes

$$\begin{aligned} \theta &\leftarrow \theta + \alpha [r_t + \gamma V_{\theta}(s_{t+1}) - V_{\theta}(s_t)] \nabla_{\theta} V_{\theta}(s_t) \\ &= \theta + \alpha [r_t + \gamma \theta^{\top} \psi(s_{t+1}) - \theta^{\top} \psi(s_t)] \psi(s_t) \\ &= \theta + \alpha [r_t + \gamma \theta^{\top} \psi_{t+1} - \theta^{\top} \psi_t] \psi_t = \theta + \alpha [r_t + \theta^{\top} (\gamma \psi_{t+1} - \psi_t)] \psi_t, \end{aligned}$$

where we have introduced  $\psi_t$  for  $\psi(s_t)$ .

- Similarly, in **batch mode**, our bootstrapped version of the on-policy prediction objective (1) becomes

$$L(\theta) = \sum_{t=0}^{T-1} \left( r_t + \gamma \theta^\top \psi(s_{t+1}) - \theta^\top \psi(s_t) \right)^2 = \sum_{t=0}^T \left( r_t - [\psi_t^\top - \gamma \psi_{t+1}^\top] \theta \right)^2.$$

# Least Squares TD learning (LSTD)

- Collect data:

$$y = \begin{bmatrix} r_0 \\ \vdots \\ r_{T-1} \end{bmatrix}, \quad \Psi = \begin{bmatrix} \psi_0^\top - \gamma \psi_1^\top \\ \vdots \\ \psi_{T-1}^\top - \gamma \psi_T^\top \end{bmatrix}.$$

- Loss function:

$$L(\theta) = \sum_{t=0}^{T-1} \left( r_t - [\psi_t^\top - \gamma \psi_{t+1}^\top] \theta \right)^2 = \frac{1}{N} \|y - \Psi \theta\|_F^2.$$

- Optimal solution (a.k.a. **TD fixed point** (Sutton and Barto 1998, Sec. 9.4)):

$$\theta^* = (\Psi^\top \Psi)^{-1} \Psi^\top y = \Psi^\dagger y.$$

## Recall: Linear regression

- Model:  $\hat{y} = f_\Theta(x) = \Theta^\top x$ .
- Loss function  $L(\Theta)$  (MSE):

$$\frac{1}{N} \sum_{k=1}^N \|\Theta^\top x_k - y_k\|_2^2 = \frac{1}{N} \|X\Theta - Y\|_F^2.$$

- Batch data:

$$X = \begin{pmatrix} x_1^\top \\ \vdots \\ x_N^\top \end{pmatrix}, \quad Y = \begin{pmatrix} y_1^\top \\ \vdots \\ y_N^\top \end{pmatrix}$$

- Optimal solution ( $\Theta^*$ ):

$$\Theta^* = (X^\top X)^{-1} X^\top Y = X^\dagger Y.$$



- Estimated value function:

$$V_{\theta^*}(s) = \theta^{*\top} \psi(s) = \sum_{k=1}^d \theta_k^* \psi_k(s).$$

## Summary / what you have learned

# Summary / what you have learned

- To cover large or continuous state spaces **function approximation** is required.
- **On-policy prediction** is straightforward with function approximation:
  - Transfer of incremental learning from the tabular case to **gradient descent** for  $\theta$ .
  - **Stochastic gradient descent** allows step-by-step based updates.
- Gradient-based prediction is **not risk free** (especially non-linear case):
  - challenging optimization problems (descent directions, learning rates, etc.),
  - local optima vs. global optimum.
- If **bootstrapping** is applied, the update target depends on  $\theta$ .
  - True gradient becomes computationally more complex.
  - **Semi-gradient** methods reduce computational burden at accuracy costs.
- **Batch learning** is often more efficient for training.

- **Linear models** can be trained in a single step using the pseudo-inverse.

# References

Sutton, R. S., and A. G. Barto. 1998. *Reinforcement Learning: An Introduction*. MIT Press.